03. Exploratory Data Analysis

What the notebook inspected, what the outputs establish, and what EDA should add.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

EDA actually performed

- Loaded the CSV and inspected df.head(), df.shape and missing-value counts.
- Created Age, Customer_Tenure_Days, Total_Spending and Total_Children.
- Generated a pairplot for Income, Recency, Response, Age, Total_Spending and Total_Children.
- Removed records where Age was 90 or above and Income was 600,000 or above; row count fell from 2,240 to 2,236.
- Calculated a numeric correlation matrix and plotted a heatmap.

Observed facts

Finding	Value / interpretation
Missing Income	24 rows initially; imputed with median
Rows filtered	4 rows removed by the two stated conditions
Retained rows	2,236
Feature matrix	18 columns after one-hot encoding
PCA variance	PC1 23.16%, PC2 11.39%, PC3 10.41%; total 44.95%

How to interpret the EDA

The pairplot and correlation heatmap are exploratory, not proof of causality. Correlations among total spending and purchase counts are expected because these features describe related aspects of customer activity. A 3D PCA plot compresses the feature space and can hide structure present in the original 18 dimensions.

Recommended additional checks

- Check duplicate IDs and duplicate rows before clustering.
- Inspect distributions and extreme values for Income and each Mnt_* category before aggregation.
- Check the six spending categories individually; summing them removes product-mix information.
- Compare channel counts and web visits; a high number of web visits with few purchases may mean something different from high web purchases.
- Check segment proportions and feature distributions by age, education and living arrangement after clustering.
- Confirm that all category levels are represented and that one-hot columns are not redundant or unstable.

Outlier treatment caution

The notebook's thresholds (Age < 90 and Income < 600,000) are hard filters, not a general outlier method. Four records were removed, but the notebook does not document a domain rationale or sensitivity analysis. Do not assume every extreme customer record is erroneous.